FIRST: Stale Rollouts Can Pick the Wrong Training Data

A Measurement-First Audit of Off-Policy Data Selection in RLVR

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Abstract

Reinforcement learning with verifiable rewards (RLVR) has to decide which prompts to train on. Fresh rollouts for every candidate are expensive, so practice ranks prompts with *stale* rollouts left over from an earlier policy, plus an importance correction. To keep variance finite, the correction reweights only one side of the trajectory: the prefix, the continuation, or a single token. Can such stale rankings be trusted? We show that the answer depends on a quantity today's pipelines never measure. In theory, any one-sided correction can flip the sign of a prompt's gradient at arbitrarily small policy divergence, and the metrics teams monitor (KL, ESS, gradient cosine) provably cannot see the flip. In practice, across Qwen2.5 7B/14B and three math tasks, what decides the outcome is how much ranking signal the prompt pool carries. We measure that signal with a split-half oracle floor, and all twelve of our conditions turn out weak: floors of 0.11–0.29 against chance 0.10. In this regime stale scores track the oracle (Fisher-combined $p = 10^{-6}$), and a free pass-rate heuristic does as well as every gradient method. We also prove two limits. Certifying a top-k choice needs 5–8 orders of magnitude more rollouts than anyone spends. And no estimator can measurably beat the floor of the oracle it is judged against. We package these findings as FIRST (Floor-Informed Reliability of SelecTion). It reads the artifacts a run already has and answers, with no new rollouts: is there signal at all, and what should you use—uniform sampling, the free pass rate, or stale scores with the disagreeing prompts sent to a small fresh audit? On our pools its pre-registered verdict is *structural absence*. The takeaway is simple: measure the floor first.

1 Introduction

RLVR pipelines choose their training data by scoring each candidate prompt with "how much would training on it help the current policy π?" and keeping the top k. Exact scores need fresh rollouts from π for every candidate, at every selection round. That is expensive. So practice reuses the rollouts and rewards produced by an earlier behavior policy β, and applies an importance correction. The correction has two axes: (a) how the current policy *reaches* a token (prefix occupancy), and (b) how it *finishes* afterwards (continuation outcome). Existing estimators correct exactly one axis. This is not an oversight: correcting both multiplies hundreds of token ratios, and the variance explodes. The implicit assumption is that the uncorrected half adds only a small bias—small enough to leave the *ranking* intact.

Published evidence already strains this assumption. The method that popularized single-token influence scoring reports, in its own appendix, a telling pair of numbers: on- and off-policy per-prompt gradients agree pointwise (cosine above 0.6 on 40 of 50 prompts), yet only 28.8% of their

*Draft v0.5 (2026-08-18). Reframed structure following the pre-registered "structural absence" branch; numbers are from completed runs (v1 gate + v2 corrected, 2 seeds); pending seed harvests will update Sections 5 and 8 only.

top-10% neighborhood structure survives [CROPI]. We treat this as a symptom, not as proof of our mechanism. The number is not decomposed over the two correction axes, and other explanations—validation noise, difficulty clustering, score calibration—can damage rank structure too. This paper supplies what is missing: the mechanism class that produces exactly this dissociation, the decomposition that makes the explanations testable against each other, and a measured map of where the failure does and does not appear.

We audit the assumption end to end. The situation turns out worse, and more structural, than a bias–variance story suggests. Our claims come in three tiers:

1. **Possibility (proved).** One-sided corrections can flip the sign of a prompt's gradient at arbitrarily small policy divergence (Section 3). Group normalization does not fix the sign. The metrics practitioners monitor—KL, ESS, gradient cosine—provably cannot detect it. One warning sign survives on stale data alone: the two one-sided cells disagreeing with each other.

2. **Regime map (measured).** Whether the flip actually happens depends on the task–capability regime (Section 5). We chart it with a pre-registered protocol and find three regimes: one where one-sided selection measurably loses precision, one where stale selection tracks the oracle, and one where selection degenerates outright.

3. **Impossibility (proved + measured).** Where selection matters most, neither the decision nor its evaluation can be secured at a practical budget. Boundary certification needs 5–8 orders of magnitude more rollouts than anyone spends (Section 6). And a measurement ceiling caps what any estimator can even *demonstrate* against a noisy oracle (Section 4).

Figure 1 summarizes the framework and how the sections fit together.

The paper's central empirical object is not an estimator but a measurement: the *split-half oracle floor*, the top-k agreement between two independent halves of the oracle's own rollouts. The floor measures how much prompt-ranking signal is recoverable at a given rollout budget; our measurement ceiling result (Proposition 2) makes it the quantity that decides whether comparative claims about selection methods are meaningful at all. Applying the same standard to ourselves, we ran FIRST (Floor-Informed Reliability of SelecTion), our pre-registered, artifact-only diagnosis, on our prompt pools (Section 8) and obtained the verdict *structural absence*: on saturated or tie-degenerate pools, no increase of the oracle budget up to K=256 rollouts per prompt is predicted to create a signal regime. That verdict, rather than a new estimator, is the paper's empirical headline—and, we argue, the more useful contribution for practitioners deciding where their fresh-rollout budget should go.

A final theme runs through the experimental sections. Selection evaluation is easy to get silently wrong in the optimistic direction: over the course of this project our own pipeline produced—and then caught—four distinct evaluation artifacts, each of which had initially produced a publishable-looking positive result (Section 7). We document each mechanism and its guard, because we believe they are endemic to the growing literature on rollout reuse and data selection for RLVR.

2 Related work

Off-policy influence and data selection. Influence scores built on a single token's importance ratio rank prompts from offline trajectories [CROPI]; in our decomposition this is g_{00} (neither axis restored). Its own appendix already shows the symptom that motivates our estimand: high pointwise gradient cosine (40/50 prompts above 0.6) coexisting with only 28.8% top-10% neighborhood preservation. Corrections that reweight the visited prefix [A Step Back, Cumulative] restore occupancy

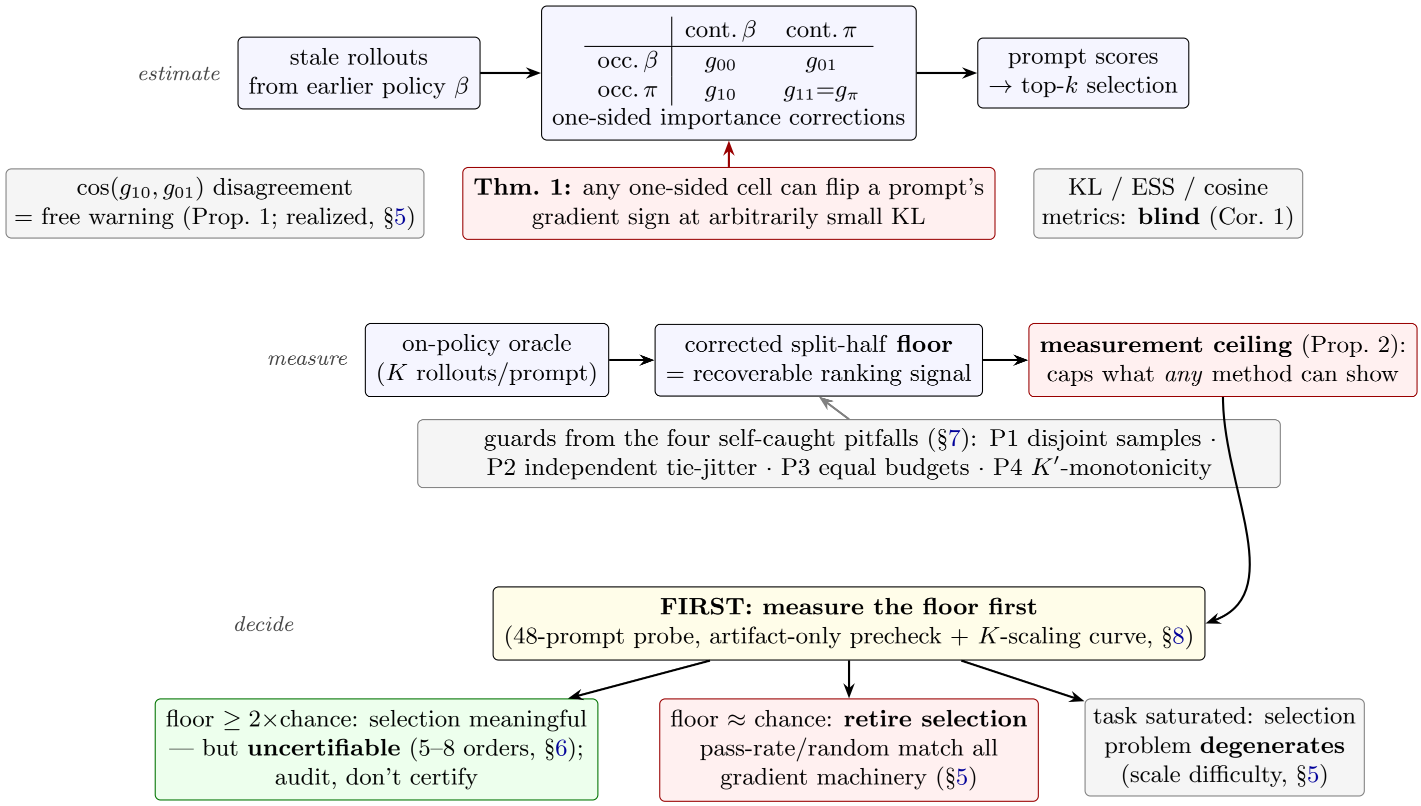

Figure 1: Framework overview. *Estimate:* stale-rollout training-data ranking uses one of four corrections in a 2×2 decomposition; one-sided cells can silently reverse rankings and the monitored metrics cannot see it (§3). *Measure:* the split-half floor of the oracle, computed under four guards, quantifies recoverable signal and induces a ceiling on what any method can demonstrate (§4). *Decide:* an artifact-only, pre-registered diagnosis routes each pool to one of three regimes; in ours the verdict was structural absence (§8).

(g_{10}); their guarantees assume a target-policy advantage that practical critic-free objectives replace by a behavior-generated outcome reward—precisely the gap our counterexamples exploit. Corrections that reweight the sampled continuation [Multi-Step] restore the outcome axis (g_{01}) but not occupancy when used for per-prompt gradient ranking. Full-trajectory corrections [TIC-GRPO] are exact but variance-explosive on long responses; they serve as the cost anchor, not a practical selector. The same tension—valuation scores failing to beat random selection until the selector is redesigned—has played out for Data Shapley in supervised data selection [Rethinking-Shapley, NASH]; our protocol (§8.1) is the RLVR analogue, with the repair driven by a reliability measurement rather than a new estimator.

Rollout allocation and learning stability. Allocation methods [GradAlign, LearnAlign] decide where to spend fresh rollouts to maximize training yield, and stale-data stabilizers keep *learning* well-behaved under reuse; neither certifies, or even measures, the correctness of the top-k *selection decision*, which is the object practitioners actually consume. Classical off-policy correction and subset selection give the statistical backdrop: conditional importance sampling [Rowland et al.], stationary-distribution correction [Liu et al.], doubly-robust policy gradients [Huang & Jiang], and information-complexity lower bounds for top-k identification [Kaufmann et al.]. Our certification impossibility (Section 6) is an empirical instance of the latter's warning: when value gaps at the boundary vanish, identification cost diverges. Both failure ingredients we study also surface in adjacent post-training literature: in on-policy distillation, token-level agreement can be driven to near-zero divergence by degenerate repetition loops while outputs collapse—a pointwise metric blind to the consumed object—and raw token log-ratio signals are dominated by the extreme tail (the top 1% of tokens carrying roughly half the gradient), the same heavy-tail pathology we measure as 0.1% ESS for the full correction [TIDE].

Reliability of noisy rankings. Our floor and ceiling analysis imports split-half reliability and the Spearman–Brown prophecy [Spearman, Brown] into RLVR selection evaluation. To our knowledge, no prior work in this literature reports the reliability of its own oracle, and Section 4 shows several of our initially-promising comparisons dissolve once it is reported.

3 Theory: one-sided corrections and what the monitored metrics cannot see

3.1 A 2×2 decomposition of stale-gradient estimators

For prompt x, response $Y = (A_1, \dots, A_T)$ with terminal verifiable reward $R(Y)$, the current-policy gradient is $g_\pi(x) = \sum_t \mathbb{E}_{H_t, A_t \sim \pi}[Q_t^\pi(H_t, A_t)\, z_t]$ with $z_t = \nabla_\theta \log \pi(A_t \mid H_t)$. With behavior trajectories, token ratios $r_j = \pi(A_j|H_j)/\beta(A_j|H_j)$, prefix products $P_t = \prod_{j<t} r_j$ and suffix products $S_t = \prod_{j>t} r_j$, four population quantities arise from the same stale data:

$$
\begin{aligned}
g_{00} &= \textstyle\sum_t \mathbb{E}_\beta[r_t R z_t] && \text{(occupancy } \beta\text{, continuation } \beta\text{)} \\
g_{10} &= \textstyle\sum_t \mathbb{E}_\beta[P_t r_t R z_t] && \text{(occupancy } \pi\text{, continuation } \beta\text{)} \\
g_{01} &= \textstyle\sum_t \mathbb{E}_\beta[r_t S_t R z_t] && \text{(occupancy } \beta\text{, continuation } \pi\text{)} \\
g_{11} &= \textstyle\sum_t \mathbb{E}_\beta[P_t r_t S_t R z_t] = g_\pi.
\end{aligned}
$$

The subscripts state which axis is restored. The exact error decompositions

$$
g_{10} - g_\pi = \sum_t \mathbb{E}_{d_\pi, \pi}\big[(Q_t^\beta - Q_t^\pi) z_t\big] \quad \text{(continuation bias)}, \qquad g_{01} - g_\pi = \sum_t \big(\mathbb{E}_{d_\beta, \pi} - \mathbb{E}_{d_\pi, \pi}\big)\big[Q_t^\pi z_t\big] \quad \text{(occupancy bias)}
$$

show that the two one-sided estimators remove *different* biases; they are not substitutes.

3.2 Sign reversal at arbitrarily small divergence

Theorem 1 (One-sided sign reversal). *For every $\epsilon > 0$ there exist two-token environments and policy pairs (π, β) with trajectory KL $O(\epsilon^2)$ such that $g_\pi = -\epsilon/2$ while $g_{00} = g_{10} = +\epsilon/2$ (continuation counterexample), and likewise $g_{00} = g_{01} = +\epsilon/2$ (occupancy counterexample). Group normalization with any group size $K \geq 2$ preserves the reversal.*

The two constructions are given in Appendix A; both flip a Bernoulli success profile so that the policies agree wherever the retained correction looks and disagree exactly where the discarded half lives. Both are machine-checked (raw gradients, exact enumeration of group normalization for $K \in \{2, 4, 8\}$ and $\epsilon \in \{0.4, 0.1, 0.01\}$) by a self-contained verification script shipped with the code.

Corollary 1 (Metric blindness). *No bound on trajectory KL, no lower bound on ESS, and no norm-blind, scale-invariant single-estimator diagnostic (e.g. pointwise cosine of the estimator against* *itself across checkpoints) can exclude sign reversal of a prompt's selection score: in the constructions above all such quantities are driven to their benign limits while the ranking direction is reversed. A gradient-norm floor or a selection-margin bound is necessary; and since $\epsilon \to 0$ also drives the absolute gradient to zero, normalized cosines conceal exactly the missing signal floor.*

Proposition 1 (Disagreement accompanies double failure)**.** *In the continuation counterexample,* $\cos(g_{10}, g_{01}) = -1$*: the two one-sided cells disagree maximally.*

Empirically—an observation, not a theorem—in a seeded random search over 5×10^4 two-token, two-parameter environments we found *zero* instances of a double reversal (both one-sided estimators reversed against g_π) accompanied by cell agreement ($\cos(g_{10}, g_{01}) > 0.9$); the identity $g_{11} \equiv g_\pi$ was verified on every sample.

Proposition 1 is deliberately modest: absence of disagreement is *not* proven to imply safety in general architectures with parameter sharing, disagreement cannot certify *which* cell is right, and its practical resolution is limited by the same margin collapse quantified in Section 6. We therefore use cell disagreement only as an *unreliability signal*—a cheap, stale-computable reason to distrust a ranking—never as a selector. The mechanism is transparent from the error decompositions: g_{10} and g_{01} subtract different bias terms from the same truth, so a large simultaneous failure of both requires the two bias terms to agree, which the decomposition makes non-generic.

4 Measuring selection: floor, ceiling, and when comparison is meaningful

Estimand. All experiments target the decision practitioners consume: the top-k set by $s_i = \cos(\mu_i, v)$, where μ_i is the prompt's mean current-policy gradient and v a validation mean gradient. We report top-k precision against an on-policy oracle, boundary margins, and rollout budgets. Pointwise cosine averages are diagnostics, never headlines.

The split-half floor. The oracle itself is a K-rollout Monte Carlo estimate. We therefore report, for every run, the *corrected split-half floor*: rollouts are split into two disjoint halves (odd/even micro-groups), each half induces a top-k set (ties broken by *independent* per-half jitter; Section 7 explains why this correction matters), and the floor is the mean overlap fraction over 20 jitter pairs. Chance is k/n. The floor measures recoverable ranking signal at budget K; it is conservative (an estimator can measurably exceed it), and Section 7 shows it is itself vulnerable to tie artifacts unless computed exactly this way.

A measurement ceiling. The floor does more than describe the oracle: it bounds what any comparison against that oracle can show.

Proposition 2 (Measurement ceiling)**.** *Model prompt scores as exchangeable Gaussians: true value t_i, half-oracle scores a_i, b_i with $\mathrm{corr}(a, b) = \rho_h$, full-$K$ oracle score o_i with reliability $\rho_f = 2\rho_h/(1+\rho_h)$ (Spearman–Brown). Then for* any *estimator s computed independently of the oracle's sampling noise, the expected top-k overlap between s and the K-rollout oracle is at most that of the best possible estimator $s = t$, namely $\mathrm{Ovl}\big(\sqrt{\rho_f}\big)$, where $\mathrm{Ovl}(c)$ is the expected top-k overlap of two Gaussian scores with correlation c. The observable floor equals $\mathrm{Ovl}(\rho_h)$, so the ceiling is a computable, monotone function of the observed floor.*

Table 1 gives the mapping at our experimental geometry (n=512, k=51), validated by simulation (Appendix B). Two consequences organize everything that follows. First, when the floor is near chance

Table 1: Floor → ceiling mapping under Proposition 2 (n=512, k=51, chance 0.100; simulation, 600–800 replicates/cell). The v2 weak-signal regime (floors 0.137–0.176) caps *any* method's measurable precision at ≈0.30–0.37; the observed stale baseline already sits there.

observed floor	0.104	0.113	0.130	0.139	0.170	0.262	0.383
half-reliability ρ_h	0.02	0.05	0.10	0.12	0.20	0.40	0.60
ceiling on measurable precision	0.170	0.219	0.278	0.300	0.372	0.516	0.640

the ceiling collapses toward chance: *under the proposition's model, no estimator can measurably win, and comparative evaluation of selection methods is void.* Second, an estimator observed *at* the ceiling has plausibly exhausted the measurable signal—further method development cannot show gains without first raising the floor (more oracle rollouts, or a pool with more signal).

Retention and its domain. When comparing regimes we report retention (prec $- c$)/(floor $- c$) with c the chance level, declared *undefined* whenever floor $- c \leq 0.02$; saturated cells otherwise produce sign-flipping or exploding ratios (the 14B GSM8K cell below is the cautionary case).

5 The regime map: where the flip does and does not happen

Setup. Qwen2.5-7B/14B-Instruct; GSM8K, MATH-500, and DAPO-Math-17k; drift between β and π is controlled by LoRA RFT for 50–400 steps; gradients are JL-projected; oracle uses K=32 fresh rollouts per prompt (8 micro-groups × 4); pools are n=256 (v1) or n=512 (v2) with k=10%. The v2 protocol was pre-registered after a full audit (equal per-cell budgets, seeded tie-breaks, independent jitter, manifests); judgment criteria were fixed before results were read. Two v2 seeds per task have completed at the time of writing; the decision rule is frozen, and the verdict will be recomputed under it when the remaining seeds arrive.

5.1 Where signal exists, one-sided selection can lose it

The v1 gate (7B GSM8K) passed its pre-registered criteria: one-sided estimators lose top-10% precision against the uncorrected baseline, and at the largest drift (400 LoRA steps) the prefix-corrected estimator collapses to precision 0.040 (overlap 1/25; exact hypergeometric $P(X \leq 1) = 0.27$, so we flag this as a directional observation, not an individually significant below-chance claim). A transparency note: the v1 floors were originally computed with the shared-jitter estimator later diagnosed as pitfall P2, which had inflated them to 0.32; the corrected re-estimation (Table 5) puts the drift-400 floor at 0.206—still the 2×-chance signal regime in which the 0.040 collapse sits far below both floor and chance; even the regime most favorable to our thesis had to survive our own guards. Downstream, fine-tuned accuracy after selection (base 0.740) spans 0.820–0.880 across oracle-, one-sided-, and random-selected pools with no consistent ordering—in this saturation-adjacent regime misranking has little room to propagate downstream, foreshadowing what follows. On 7B MATH-500 (corrected floor 0.288, the strongest and cleanest signal regime: a *rising* K'-curve, Table 5) the *ordering flips*: the uncorrected g_{00} is worst (0.160) and every stale estimator retains only 0.28–0.64 of the available signal; at 14B MATH-500 the loss concentrates on the suffix axis (signal retention g_{01}=0.44 vs. g_{10}=1.00), though its declining K'-curve counsels caution on the absolute floor there. Which cell fails is regime-dependent; *that some cell fails while every monitored metric looks normal* is the invariant, exactly as Theorem 1 licenses and no more.

Table 2: v2 corrected results (n=512, k=51, chance 0.100; 2 seeds). Floors are corrected split-half (independent jitter). DAPO precisions are jitter-sensitive (±0.08) due to margin-0 tie degeneracy and shown as bands.

run	floor	g_{00}	g_{10}	g_{01}	g_{11}	regime
GSM8K s1	0.137	0.294	0.275	0.196	0.196	weak signal; stale at ceiling
GSM8K s2	0.176	0.235	0.196	0.216	0.216	weak signal
DAPO s1	0.176	0.04–0.16 (tie-degenerate)				margin 0.0; chance
DAPO s2	0.176	0.04–0.16 (tie-degenerate)				margin 0.0; chance

Table 3: Prompt-level sign reversal against the fresh oracle (completed v2 GSM8K seeds; zero-signal prompts excluded, denominators shown). Raw rates must be read against the anchor row—the oracle's split-half self-disagreement on the same prompts—not against zero. Decision band = oracle ranks $k \pm 26$.

	overall reversal		decision-band reversal	
	s1	s2	s1	s2
g_{00}	67/163	80/168	12/37	12/34
g_{10}	68/163	73/168	13/37	10/34
g_{01}	61/163	63/168	10/37	8/34
g_{11} (full)	59/163	64/168	9/37	8/34
oracle self-disagreement (anchor)	66/168	77/180	10/37	7/35

5.2 Where signal is weak, stale selection tracks the oracle

The corrected v2 runs put every completed condition at floors 0.137–0.176 against chance 0.100 (Table 2). Here stale scores are *not* distinguishable from the oracle—in fact they track it: on GSM8K the uncorrected stale baseline achieves precision 0.294/0.235 (seeds 1/2), upper-tail exact $p = 2.8 \times 10^{-5}$ and 2.1×10^{-3}, Fisher-combined $p = 1.0 \times 10^{-6}$ (the two seeds share the prompt pool; conditional on the pool, the hypergeometric nulls are driven by independent rollout randomness, which is what the combination assumes). Crucially, 0.294 is *at the measurement ceiling* (≈0.30 at floor 0.137; Table 1): the stale estimator has exhausted what the oracle can resolve, so neither its one-sided competitors' losses nor any method's gains are measurable in this regime. The DAPO pool is tie-degenerate (boundary margins exactly 0.0; precision itself moves by ±0.08 under tie-break jitter, so we report bands): all stale estimators sit at chance, and so does everything else.

[PENDING — v2 seed 3, running] Additional GSM8K/DAPO seeds are being harvested; Tables 2 and 5 will gain one row group each and all seed-pooled statistics will be recomputed under the pre-registered rules. Verdict logic is frozen; only numbers change. *Reserved space: half a page (updated tables + two sentences).*

Prompt-level reversal prevalence (preliminary). Top-k overlap is a coarse summary; the natural reviewer question—*how often does the sign actually flip?*—has a finer-grained, prompt-level answer. Table 3 re-aggregates the completed v2 GSM8K seeds: stale scores disagree in sign with the fresh oracle on 36–48% of signal-bearing prompts overall, 24–35% inside the decision band, and 6–15 of the oracle's top-51 prompts receive a sign-flipped stale score. The anchor row calibrates these rates: an independent half of the oracle disagrees in sign with the other half on 39%/43% of the same prompts (decision band: 27%/20%). Overall reversal for every cell sits within a few points of this anchor, so the headline rates are dominated by oracle noise and we do not read them as estimator failure; the information is in two contrasts that place the same oracle noise on both sides. First, inside the decision band the one-sided cells sit above the anchor while the full correction sits on it: pooled band reversal is 0.34 (g_{00}) and 0.32 (g_{10}) against 0.24 for g_{11} and 0.24 for the anchor, with g_{01} indistinguishable from full (0.25); prompt-paired McNemar tests against g_{11} (exact, over all signal-bearing prompts) reach significance for g_{00} in seed 2 (27 vs. 11 discordant prompts, $p = 0.014$), with the remaining paired contrasts not significant ($p \geq 0.13$)—pooling is over a shared prompt pool, so we present this as replication across seeds, not independent evidence. Notably, this resolution exceeds what top-k precision offers in the same regime (Table 2), where the ceiling hides any separation. Second, the theory's warning sign operates at the prompt level: conditioned on the two one-sided cells disagreeing in sign, the prefix-corrected cell g_{10} is the reversed one in 59%/62% of cases (seeds 1/2) against a 37% reversal rate under agreement (Fisher exact $p = 0.015/0.007$), and the same pattern replicates in the strong-signal v1 condition (7B, MATH-500): 81% against 45%, $p = 0.012$. In the saturated 14B regime, where selection itself degenerates (§5.3), the alarm is correspondingly uninformative ($p \geq 0.44$)—as it should be, since there is no signal left to protect. This is support from real LLM runs for the disagreement signal of Proposition 1, with one estimand caveat stated explicitly: for scalar per-prompt scores a *double* reversal manifests as cell *agreement*, so the 37–45% under agreement is exactly the alarm's blind spot—consistent with "agreement guarantees nothing," and distinct from the proposition's vector-cosine statement. The DAPO seeds contribute 2–4 signal-bearing prompts each (tie degeneracy, §5) and are excluded.

5.3 Where capability saturates, selection degenerates

At 14B on GSM8K (∼95% accuracy) the pool contains 132/256 zero-reward-variance prompts; every estimator's precision equals the chance level (0.087–0.111 vs. 0.098), and the oracle disagrees *with itself* below chance (floor $0.080 < 0.098$). Selection, ranking, and certification all degenerate together: no method, and no certificate, can matter here. The practical corollary is that selection pipelines must scale candidate difficulty with model capability, or silently pay for machinery that cannot change the outcome.

6 Certification is out of reach where it matters

Even where a point estimate selects well, one may ask for the decision to be *certified*: spend fresh rollouts near the ranking boundary until confidence intervals separate the top-k set. We implemented this (sequential boundary refinement with shared-validation error propagation; pseudocode in Appendix C)—not as a method contribution, but as a measuring instrument. The instrument returns a structural negative. Across every completed condition (both scales, three tasks, drift 50–400, selection fractions 5–25%), certification fails at budgets 1.02–1.04× the uncorrected cost. The reason is geometric: top-k boundary margins are 0.00–0.24°, while achievable confidence radii are 51–180°. The gap is not a constant factor: with radius $\alpha_v(m) = \arcsin(c/\sqrt{m})$, reaching the observed margins requires 4.1×10^5 to 1.0×10^8 times more observations—5–8 orders of magnitude in budget, robust to (ϵ, δ)-PAC slack and deepened validation. Prompt values are intrinsically dense at the selection boundary: *selection decisions resist exact certification precisely where selection matters—at every margin we observed.* Approximate-membership and indifference-zone formulations are weaker demands our measurements do not exclude. Fresh rollouts spent on exact certification bought nothing in any condition we measured; Section 9 argues they belong in floor measurement instead.

7 Four evaluation pitfalls our own pipeline caught

Every mechanism below produced, at some point in this project, a clean positive result that survived a first review and failed under a targeted audit. We report them as first-class findings; each comes with a guard now built into our protocol, and we suspect instances of each exist in the current rollout-reuse literature.

P1: Oracle-sample reuse (leakage). Our v1 hybrid intervention (completing the "missing half" with fresh continuations) showed a perfect 21/21-condition recovery, mean +0.46/+0.49 precision—the would-be causal centerpiece. A code audit found the treated cell reused rollouts that also defined the oracle; the shared noise, not causation, produced the recovery. A pre-registered equal-budget decisive experiment with disjoint samples showed *no consistent recovery* (e.g. GSM8K s1: fully-fresh cell 0.38 *below* mixed cell 0.46). *Guard:* strict sample disjointness between treatment and oracle (odd/even micro-group separation), enforced structurally.

P2: Shared-jitter tie inflation. Our first floor implementation broke score ties with a jitter shared across the two halves. On a tie-degenerate pool (DAPO, margins 0.0) this inflated the floor from 0.176 to 0.804—which briefly supported a striking (and wrong) "inverse correlation" conclusion—and it had quietly inflated the v1 floors too (0.32 reported vs. 0.15–0.21 corrected), which means every absolute floor in this literature computed with tie-unaware agreement warrants re-examination. *Guard:* independent per-half jitter streams, 20 seeded pairs, plus reporting precision *bands* under jitter on tie-heavy pools.

P3: Cell-budget imbalance. Comparing estimator cells with unequal effective rollout counts silently favors the better-funded cell; our v1 hybrid grid had this flaw on top of P1. *Guard:* equal-K redesign of all cell comparisons, verified from manifests.

P4: Saturation-shared overlap inflation. An above-chance floor can be an artifact of *shared liveness* rather than shared ranking: prompts that saturate in both halves (all-correct or all-wrong) drop to an exactly-tied score, and the surviving "live" sets overlap between halves, producing top-k agreement with zero true ranking signal. In a synthetic pool with *no* ranking signal, saturation sharing alone yields floor 0.126 at K'=8, decaying toward chance (0.110) at K'=32—the signature being a floor that *decreases* as oracle budget grows. Our measured K-scaling curves (Section 8) show exactly this decreasing shape, which means even our reported weak-signal floors are best read as upper bounds. *Guard:* always report the floor as a function of K'; treat a non-increasing floor curve as artifact evidence, and report tie mass alongside.

8 FIRST: diagnosing signal absence from existing artifacts

The pitfalls motivate the final contribution: FIRST, a pre-registered, artifact-only procedure (no new rollouts) that decides whether a pool supports selection *before* any new experiment is funded. It has three stages, all computed from artifacts every RLVR selection run already produces.

Table 4: K-scaling floor curves (observed exact recomputation; prediction by Spearman–Brown + Gaussian overlap). Chance 0.100. Note the *decreasing* observed curves—the P4 artifact signature—and the near-zero half-reliabilities. Verdict (pre-registered): structural absence.

run	r_1	K'=8	16	24	32	pred. 128	pred. 256
GSM8K s1	0.045	0.263	0.208	0.201	0.182	0.288	0.391
GSM8K s2	−0.005	0.220	0.182	0.170	0.156	0.094	0.094
DAPO s1	0.009	0.112	0.123	0.139	0.146	0.147	0.182
DAPO s2	−0.038	0.119	0.148	0.158	0.158	0.094	0.094

Stage 1: conditional-subset precheck. Any proposed pool intervention definable as a filter on existing prompts (e.g. "keep only live prompts, $0 <$ pass-rate < 1") already exists as a subset of completed runs. Computing the corrected conditional floor on that subset (with bootstrap CIs) prechecks the intervention at zero GPU cost: if the conditional floor does not clear $2\times$ chance in a majority of eligible runs, building the new pool has no evidential basis. On our pools this returned NO-GO, and the intervention (a live-only GSM8K pool) was cancelled without spending a GPU-hour.

Stage 2: K-scaling floor curve. From stored micro-group gradients the floor is *exactly* recomputed at K'=8, 16, 24, 32, and extrapolated to K'=64–256 by Spearman–Brown on the half-reliability r_1 mapped through the Gaussian overlap function, with a calibration check at the largest observed K'. The pre-registered verdict rule: if a majority of eligible runs are predicted to reach $2\times$ chance by $K' \leq 256$, recommend oracle expansion; otherwise declare structural absence.

Table 4 shows the result: one GSM8K seed extrapolates to $2\times$ chance at K'=128 from r_1=0.045; the other is flat at chance; both DAPO runs never reach it. Majority not met: *structural absence*—meaning no recoverable ranking signal *within the evaluated pools and budget range*. This is an epistemic verdict about what these pools and budgets can support, not an ontological claim that prompt-value differences do not exist.

Running the same analysis over every artifact-complete run in the project (twelve conditions including the v1 gate family; Table 5) widens the evidence base without touching the pre-registered vote: signal regimes are rare (2–3 of 12), the two credible ones are exactly where the regime map's one-sided losses live (7B MATH-500 and maximum-drift GSM8K), and half of all observed curves *decline* in K'—the P4 artifact signature—so absolute floors elsewhere are best read as upper bounds.

Stage 3 (replay): the ceiling made visible in budget space. As a final check we replayed all fresh-budget allocation policies on the completed v2 runs against a sample-disjoint truth (odd micro-groups only): stale cells, a pass-rate heuristic, random, uniform and boundary audits at 1–25% budgets, a sequential disagreement-guided audit, full-fresh at up to 100% budget, and our own pre-registered floor-gated protocol. The result is the measurement ceiling made concrete. On weak-signal GSM8K no fresh budget yields an improvement: full-fresh at 100% budget scores *lower* (0.167) than the zero-cost stale baseline (0.235), the floor-gated protocol loses (0.172)—by its pre-registered loss condition it is retired to the diagnostic role this section describes—and the best zero-cost policy is not a gradient method at all but the behavior pass-rate heuristic (0.255). On DAPO everything lands at chance (0.08–0.16, truth margins 0.0). Meanwhile the monitored metrics stay in their normal ranges, precisely as Corollary 1 warns, and the full correction's effective sample size is 0.1% (clip fraction 0.94–1.00)—the variance-side reason one-sided truncations exist in the

Table 5: Extended K-scaling evidence over all twelve completed conditions (corrected floors; "shape" is the observed K'=8→32 direction, with ↓ marking the P4 artifact signature). The pre-registered verdict is computed only on the v2 rows and is unchanged; the extension shows how rare signal regimes are: three conditions clear 2× chance at K'=32, and only two of those with a rising curve. Floors here use random half-assignment resampling and may differ slightly from the fixed odd/even-split floors of Table 2.

condition	task	r_1	floor @ K'=32	reaches 2× chance	shape
v1 drift-50	GSM8K	0.020	0.177	pred. K'=256	↑
v1 drift-100	GSM8K	0.053	0.146	pred. K'=128	↓
v1 drift-200	GSM8K	0.067	0.147	pred. K'=64	↓
v1 drift-400	GSM8K	0.065	**0.206**	at K'=32	↑
v1 14B	GSM8K	0.018	0.113	never	↓
v1 7B	MATH-500	0.070	**0.288**	at K'=32	↑
v1 14B	MATH-500	0.075	0.221	at K'=32	↓
v2 s1/s2	GSM8K	0.045/−0.005	0.182/0.156	pred. 128/never	↓↓
v2 s1/s2	DAPO	0.009/−0.038	0.146/0.158	never	↑↑
v2 s2	MATH-500	0.001	0.181	never	↓

first place, measured in the same runs that show what the truncation costs. Two pre-registered caveats accompany the method. The calibration check shows the Gaussian model under-predicts observed floors at K'=32 (e.g. 0.094 predicted vs. 0.156 observed)—consistent with the P4 artifact inflating the *observed* values, which makes the verdict conservative in the safe direction. And the single positive extrapolation rests on an r_1 small enough to be within the P4 artifact's reach, so we treat it as motivation for a cheap re-check when further seeds land, not as grounds for a GPU expansion.

8.1 From diagnosis to decision: the FIRST protocol

The verdict is not the end of the pipeline. It is the first branch of a decision. Each regime in the map comes with a selection policy, and our own replays already price each one. So the diagnosis composes into a protocol. The data-valuation literature made the same move for supervised learning: when Shapley-based selection proved no better than random, the answer was a repaired selector, not a retreat [Rethinking-Shapley, NASH].

The FIRST protocol inputs: the behavior rollouts the run already has (optionally a 48-prompt fresh probe); fix the threshold $\tau = 2\times$chance *before* looking.

1. **Measure.** Compute the corrected split-half floor (independent jitter) and the K-scaling curve with its Spearman–Brown budget estimate K^* (§8); run the four pitfall guards (§7).
2. **Branch.**
   (a) *No reachable signal* (floor $\approx$ chance, no finite K^*): abstain—sample uniformly and spend nothing on scoring. Every estimator we measured sits at chance here, and the oracle disagrees with itself (§5.3).
   (b) *Weak signal* (chance $<$ floor $< \tau$): rank by the free behavior pass rate. In our replays it matched or beat every gradient estimator (0.255 vs. 0.235 best stale), and the measurement ceiling shows nothing can *measurably* beat it in this regime (§4).
   (c) *Measurable signal* (floor $\geq \tau$): stale scores are usable. Compute both one-sided cells anyway, and send the prompts where they disagree in sign to a fresh audit (or drop them). Under disagreement the reversal rate was 59–81%; under agreement, 37–45% (three runs, $p \leq 0.015$). Agreement guarantees nothing—that is the stated blind spot.
3. **Report.** Attach the floor, the ceiling, and the branch taken to any downstream selection claim; a result above its own oracle's ceiling is an artifact by Proposition 2.

The verdict costs no new rollouts; branch (c)'s audit spends fresh budget only on the disagreement set, which in our runs was 14–25% of signal-bearing prompts.

[PENDING — capability–difficulty sweep (decision gate 9/10)] A small-model sweep (Qwen3.5 0.8B/2B/4B on the same pools) would chart the floor as a function of the capability–difficulty gap—the one controllable knob that could *create* a signal regime and turn the regime map into a causal phase diagram. If run, it adds one figure (floor phase map) and one paragraph here; if not, the map remains observational, as stated in Limitations. *Reserved space: one figure + half a page.*

9 Practical guidance, limitations, and outlook

For practitioners. (1) *Measure the floor first*—run FIRST and take the branch its protocol prescribes (§8.1). A 48-prompt fresh probe is enough to estimate the split-half floor. Below $2\times$ chance (fix the threshold *before* looking), do not spend on selection. In our replays the best weak-signal policy was the free pass-rate heuristic, random was close behind, and no fresh budget up to 100% bought an improvement. (2) *Never budget for boundary certification*; the 5–8 order-of-magnitude gap is structural. (3) *Scale candidate difficulty with capability*, or selection degenerates regardless of estimator. (4) *Report floor and ceiling with any selection comparison*; a claimed improvement above the measurement ceiling of its own oracle indicates assumption violation, estimator–oracle dependence, or an evaluation artifact. (5) *Distrust one-sided rankings when the two one-sided cells disagree*; agreement guarantees nothing, but disagreement is a free warning—formalized as branch (c) of the protocol.

Limitations. Empirics are Qwen2.5 (7B/14B), math tasks, and LoRA drift; two completed v2 seeds (remaining seeds and a small-model capability–difficulty sweep are running and will be folded in; the verdict logic is pre-registered). The ceiling proposition assumes an exchangeable Gaussian score model (simulation-validated at our geometry; the analytic program of proving it distribution-free is open). The disagreement result is an existence-plus-search finding, not a general sufficiency theorem. Theory counterexamples are two-token constructions—deliberately: they show what *cannot* be excluded by the monitored metrics, not what typically happens; the regime map is the empirical complement.

Safety relevance. Data selection is a curation channel: whatever a misranked estimator systematically promotes gets amplified by subsequent training. Corollary 1 then has a safety reading—if stale selection happens to promote prompts on which shortcut or reward-hacking behavior scores well, the drift metrics a team monitors cannot flag the promotion any more than they can flag the sign reversal itself, so the selection layer is an effectively unaudited input to the reward loop. The measurement-first protocol applies unchanged when the selection target is a safety metric rather than accuracy, and we recommend running it there before trusting any stale-scored curation.

Outlook. We view the negative results as load-bearing: they delimit where the field's growing investment in stale-rollout data selection can possibly pay off (unsaturated, tie-free, signal-bearing pools—which our diagnosis identifies for free) and where it cannot. The measurement-first protocol—floor, ceiling, preregistration, and the four guards—is applicable to any selection-method paper, including ours.

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A Counterexample constructions and machine verification

Continuation counterexample (reverses g_{00}, g_{10}). Two-token MDP; first action $A_1 \sim \text{Bernoulli}(q)$, q=1/2, identical under both policies; reward $R = A_2$. Behavior success profile $Q^\beta(A_1{=}0) = \frac{1}{2} - \epsilon$, $Q^\beta(A_1{=}1) = \frac{1}{2} + \epsilon$; current policy swaps the two values. The gradient w.r.t. the logit of q is $g_\pi = -\epsilon/2$ while $g_{00} = g_{10} = +\epsilon/2$: the prefix already matches perfectly, so a perfect prefix ratio changes nothing, and the behavior-policy *ending* drives the sign. Mean suffix KL is $O(\epsilon^2)$; the 1-D cosine is -1 throughout.

Occupancy counterexample (reverses g_{00}, g_{01}). Behavior visits $P(A_1{=}0) = \frac{1}{2} + \epsilon$, current $\frac{1}{2} - \epsilon$; the second action is Bernoulli(1/2) under both, and $R = A_2$ if A_1=0, else $1 - A_2$. For the second-action logit, $g_\pi = -\epsilon/2$ while $g_{00} = g_{01} = +\epsilon/2$: no tokens remain after A_2, so the future ratio is exactly 1, and the stale *visit* distribution drives the sign. Trajectory KL is again $O(\epsilon^2)$.

Group normalization. Both constructions have overall success rate 1/2 under both policies; binary group standardization multiplies each direction by the same nonnegative scalar, hence preserves the reversal (exact enumeration for $K \in \{2, 4, 8\}$, $\epsilon \in \{0.4, 0.1, 0.01\}$ in the verification script, alongside leave-one-out unbiasedness and the confidence radius bound used in Section 6).

B Measurement ceiling: derivation sketch and validation

Under the model of Proposition 2, write the full-K oracle score as $o = \sqrt{\rho_f}\, t + \sqrt{1 - \rho_f}\, \eta$ with η independent noise. For any estimator s independent of η, $\text{corr}(s, o) = \text{corr}(s, t)\sqrt{\rho_f} \leq \sqrt{\rho_f}$, with equality iff s is (a monotone transform of) t. Expected top-k overlap between two jointly Gaussian scores is increasing in their correlation, which yields the ceiling $\text{Ovl}(\sqrt{\rho_f})$; the observable floor is $\text{Ovl}(\rho_h)$ by the same function. Simulation at n=512, k=51 (600–800 replicates per cell; independent-jitter tie handling) produces Table 1; the mapping is monotone and smooth, so inverting the observed floor through it is well-posed. The Gaussian assumption matters: heavy-tailed score distributions with a stable head can exceed the Gaussian ceiling estimate, which is precisely the P4 caveat and why we report the ceiling alongside, not instead of, the K-scaling curve.

C Sequential boundary certification (measuring instrument)

Off-policy scores serve as a prior; every candidate receives a small fresh micro-group; scores carry confidence balls (shared validation error propagated); fresh rollouts are then spent only on candidates whose balls straddle the top-k boundary, until $\min L_{\text{selected}} > \max U_{\text{rest}}$ or the budget is exhausted, in which case *certification failure is reported, never hidden.* Variants (scalar CI, (ϵ, δ)-PAC with $\epsilon \leq 0.05$) behave identically in all our conditions because the binding constraint is the boundary margin itself (Section 6).

D Preregistration and audit trail

All v2 judgment criteria, the precheck GO/NO-GO rule, the K-scaling verdict rule, and the reframing decision tree ("structural absence $\Rightarrow$ this paper's structure") were registered in the project log before the corresponding results were computed; the four pitfalls each carry a before/after commit pair. The verification script, manifests, and all readouts ship with the code release.